

Manual

Quality Management Sub System to SAP QM QMS-SQM 8.1

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1 Quality Management Sub System to SAP QM

Purpose

This component is used when inspection planning and possible evaluation is performed in SAP-QM but the inspection data are collected in HYDRA.

Implementation Considerations

Use of SAP-QM is required. This function package is useful when you combine the AIP data entry client for BDE and MDE with the collection of inspection data or SAP inspection batches. This setup provides for optimal control of appropriate inspections e.g. when reaching a time or piece interval.

Integration

The AIP data entry client is required to enter inspection data. In addition, the collected data can be used in the component "Evaluations in the QM Subsystem" to generate graphical presentations of the measured values.

Features

The central function is the presentation of the inspection batches transferred by SAP through the interface and the visualization of the inspection data and decisions that HYDRA has collected for these inspection batches. The inspection requirements (corresponding to SAP inspection batches) and the corresponding inspection steps can be related to the SAP default values and the quality decisions taken. This includes the detailed observation of inspection points (if used) and the respective inspection characteristics.

2 Catalogs

2.1 Summary

Menu	Master data → Quality management → Catalog
Transaction code	cat
Function authorization	cat

2.2 Utilization

The contents of this application cannot be changed as it includes catalog entries that are exclusively populated by SAP-QM using the QM-IDI interface. In addition, the usage decisions about inspection points, which are also used for in-production inspection (no SAP-QM or QM subsystem) are kept here. The inspection point decisions that do not derive from SAP-QM can be changed by MPDV consulting while the system is customized. This includes the modification of field names as well as the deactivation of individual entries or the creation of new inspection point decisions.

Examples for SAP-QM catalogs:

- Usage decisions on the inspection point and inspection requirement (inspection batch),
- Defect types,
- Defect locations

An overview of the selection lists that can theoretically be opened during the inspection process is provided. Consequently, the leading planning system (SAP-QM) determines which catalogs are available at what position.

2.3 Integration

The catalog contents are available in the corresponding inspection processes as selection lists, e.g. to select an inspection point decision, inspection batch decision to be uploaded to SAP-QM.

2.4 Selection Criteria

The application provides the following selection criteria:

Catalog type

Number of the catalog type

Selected set

Subgroup of the catalog type

Site

Shows to which site it refers

Code

ID number (alphanumeric)

Code group

Group ID of individual codes

Selection

Checkbox

A match code search is available for all fields, except for the filter field "selection". As regards the selection, it can be defined whether all data records are to be displayed or only those with enabled or disabled selection.

2.5 Field descriptions

As no data can be changed, the individual fields are not described in more detail. The single fields can be found in the application.

2.6 Editing functions

This application does not provide editing, insert and deletion functions.

2.7 "Catalog" Detail Application

As already mentioned, catalog entries can neither be changed nor deleted. They can only be edited using the interface (e.g. if QM-IDI is used as SAP-QM subsystem) or by MPDV customizing.

Detail data of a catalog entry is only provided in list form and divided among the tabs

- Catalog and
- Administration

The individual fields are not described in detail as data cannot be changed.

3 Machine Status Change

3.1 1.1 Summary

Menu	Master data → Quality management → QM activity, machine status change
Transaction code	mastqm
Function authorization	mastqm

This application aims at inspection planners, although inspection planning is performed in a higher-level system (e.g. SAP-QM).

As a prerequisite, the "machine status change" function has to be used at the shop floor terminal of the subsystem.

3.2 Utilization

This application allows for a definition with respect to workstations to be made specifying which machine status changes are to trigger inspections.

This function is not part of inspection planning of the higher-level system but supplements it on subsystem level by automatically generating due inspections when machine statuses change.

3.3 Selection criteria

The application provides the following selection criteria:

Workplace

Direct input (match code) or selection list of the workplace catalog including direct transfer

Source status

Direct input or selection list of the machine status

Target status

Direct input or selection list of the machine status

Active

Active, inactive or all statuses are filtered.

Generate inspection point

Configurations with or without the generation of inspection points or both types are filtered.

3.4 Field descriptions

Workplace

The workplace for which a configuration has been created is filtered.

Source status

Starts the status assignment list

Target status

Starts the status assignment list

Active

Selects e.g. active entries.

Generate inspection point

Selects the activities, for example, that are supposed to trigger inspections.

3.5 Toolbar

There are no special function buttons.

4 "QM activity, machine status change" detail application

This application is only relevant if HYDRA is used as QM subsystem, e.g. in connection with SAP-QM. Data is only displayed in list view.

This detail application allows for configurations with respect to the workstation to be made, specifying whether a status change of a machine to which an inspection step is logged on, is to trigger an activity (due inspection) as regards the check that is currently being performed, e.g. the generation of an inspection point.

Machine status changes that are caused by logging an operation on (DLG=A_AN) and by the user changing the status manually (DLG=M_MST) are only considered for the generation of inspection points.

The corresponding buttons of the toolbar allow for a configuration to be created, changed or deleted. When a new data record is created, the user is supported by the selection lists "workplace" and "status assignment". Provided that a data record is changed, the flags "active" and "generate inspection point" as well as the contents of user fields that might be defined can only be changed.

The "position" field uniquely identifies the data record and, therefore, must be unambiguous.

4.1 Field description

Position

Uniquely identifies the data record and, therefore, must be unambiguous.

Workplace

Defines the workplace for which a configuration is to be created.

Source status

Specifies the source status on the basis of which e.g. an inspection point is to be generated after switching to the machine status "x".

Target status

On the basis of the source status "x", e.g. an inspection point is to be generated after switching to the machine status that is defined here.

Active

The configuration is enabled.

Generate inspection point

If this option is activated, an inspection point is generated, once the machine status changes.

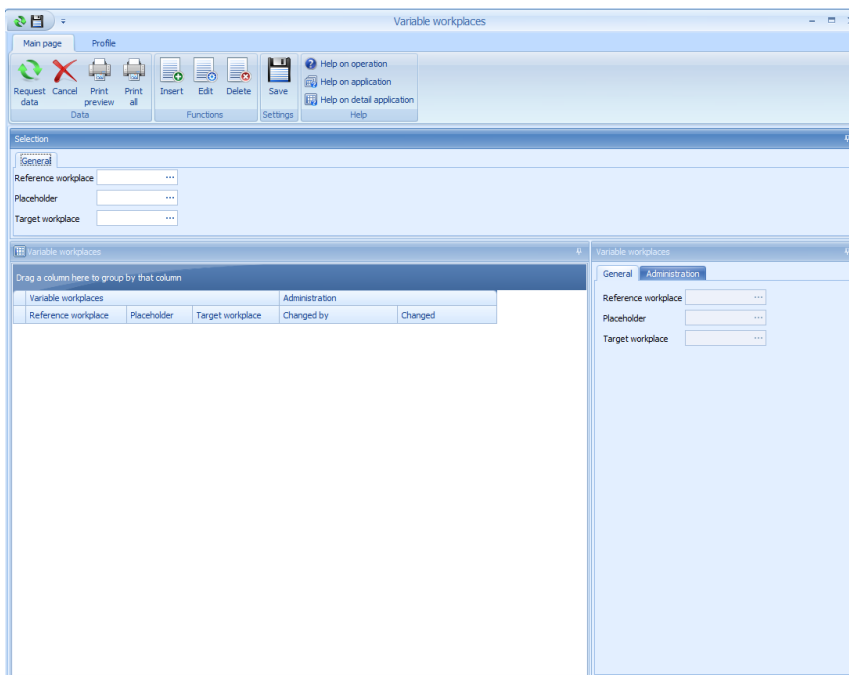
5 Variable Workplaces

Summary

Menu	Master data → Quality management → Variable workplaces
Transaction code	vawo
Function authorization	vawo

Although inspection planning is performed in a higher-level system (e.g. SAP-QM), this application addresses the group of inspection planners, as this application determines which inspections (QM operations/processes) are performed at which workplaces.

This function does not belong to inspection planning of the higher-level system but complements it on sub-system level by defining the actual inspection stations/workstations in detail.



Utilization

If linked QM operations/processes are used in SAP, for example, it cannot be planned in SAP to which workplace/machine the QM operations are to be logged on. For this reason, it has to be defined for the workstation transferred by QM-IDI (or any other ERP interface) to which productive workplace (reference workplace) it pertains and onto which QM workplace (target workplace) it is to be logged on.

This controls where the QM operations 0021 and 0025 are to be logged on along with logging the productive operation 0020 on. The QM operation 0021 is logged on, for example, on the terminal onto which the productive operation is logged on as well. The QM operation 0025, however, is logged on to the "laboratory" workplace of another shop floor terminal.

If linked QM operations/processes are used an assignment has to be established while the system is customized. Ideally, a "concept" defining the connection rules should be defined in advance. It is indispensable to customize the system according to the user's requirements to be able to implement the connection rules that are to be defined.

It is not sufficient to only use this application.

The below example describes a possible connection rule.

- The operation number links the productive operations and QM operations.
- An operation template is created for the combination of workplace and order type including the assigned processing code. In this context, it does not play a role which order is assigned.
SYSCAQ = is to be linked
CAQPUR = separate QM operation
- Creating the QM operation in HYDRA triggers the search for an OP template taking into account the workplace and the order type (derives from the QM-IDI interface). The processing code, which in turn determines whether or not the QM operation is to be linked, is determined in the template that is found.
- A productive, preceding operation is searched in the corresponding production order for an QM operation that is transferred via the interface. The operation number specifies the search order. The productive operation with the next least operation number is searched. An operation that is found in this way is entered as master operation for the QM operation.
- The application of variable workplaces defines which QM operations are to be logged on to which workplaces/machines.

If workplaces are defined as mere QM workstations in HYDRA they have to be assigned the type "CAQ inspection station" as individual workplace.

Integration

This application is only required if linked QM operations/processes are in use. For this reason, the application is restricted to being used in connection with HYDRA used as QM subsystem, e.g. with SAP-QM.

In exceptional cases, which have to be checked in each individual case, this application may also be used with HYDRA inspection planning in the HYDRA inspection planning/inspection requirements application, provided that linked QM operations are in use.

Prerequisite

Linked QM operations/processes need to be used.

Selection criteria

The application provides the following selection criteria:

Reference workplace

Direct input (match code) or selection list of the machine/workplace catalog including acceptance function.

Placeholder

Direct input (match code) or selection list of the machine/workplace catalog including acceptance function.

Target workplace

Direct input (match code) or selection list of the machine/workplace catalog including acceptance function.

Field descriptions

Reference workplace

Direct input or selection list of the machine/workplace catalog including acceptance function.

Placeholder

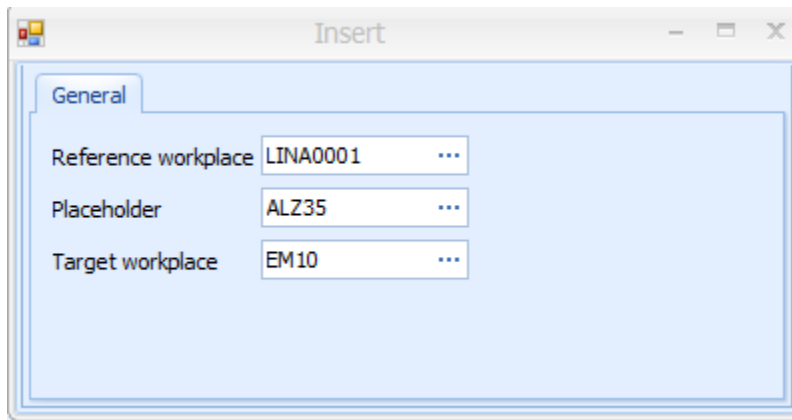
Direct input or selection list of the machine/workplace catalog including acceptance function.

Target workplace

Direct input or selection list of the machine/workplace catalog including acceptance function.

Editing functions

The following dialog opens to edit a data record:



Toolbar

The toolbar does not provide any special functions/features.

6 Overview

6.1 Notes to this Document

This document describes the application "Inspection Requirements QMS" within the Manufacturing Operation Center (MOC). General information on the use of the MOC can be found in the document ["moc_cc.pdf"](#).

7 Inspection Requirements

Target group:

HYDRA as a QMS subsystem requires inspection planning and generation of inspection batches in SAP, where collecting the inspection results and taking quality decisions (e.g. inspection batch usage decisions) are an integral part of the shop floor terminals or MOCs. Since collecting the inspection results is an integral part of the shop floor terminal, the application is aimed at the employee subgroup with the authority to take quality decisions.

Within MOC the SAP inspection batches are mapped onto inspection requirements.

Application Determination:

This document describes the SAP inspection batch as a QM subsystem within MOC. SAP is considered the leading system when planning and generating inspection batches; the quality decisions are based on data entered into the MOC and these are uploaded into SAP together with the individual results.

Within MOC the inspection requirements and the corresponding inspection steps can be related to the SAP default values and the quality decisions taken in MOC. This includes the detailed observation of inspection points (if used) and the respective inspection plan characteristics.

Boundaries

This application examines the results and SAP specifications transmitted through the QM-IDI Recording inspection results is not part of this application.

7.1 Function Calls

Menu	Inspection Requirements → Quality Assurance → QM Subsystem
Transaction code	irqms
Function authorization	irqm

7.2 Application Layout

7.3 Tool bar

The tool bar contains the function calls available to the application and the links to other applications. The functions on the General page of the tool bar are applicable to all detail applications. Additional pages contain the functions specific to the respective detail application, in addition to the standard functions Help, Request Data, Application Settings, Save and Print Preview. The individual application functions are described below.

Data Category



Request Data

The information to be displayed in the application is requested according to specified selection criteria. This processing can take considerable time to complete depending on the amount of data to be processed and displayed.



Cancel

This function is used to abort the query resulting from "Request Data".



Print Preview

This function displays the print preview of the highlighted detail application. The print preview contains additional ways to control the print result as well as functions to export the information displayed to other formats like PDF, Excel or Image files.

Detail Application Function Category "Inspection Requirements"



Complete Inspection Requirement

The SAP inspection batch results in a usage decision. The usage decision is based on the inspection results.

Detail Application Function Category "Inspection Steps"



Complete Inspection Step

Based on the inspection results the inspection step is completed.

Detail Application Function Category "Inspection Points"



Insert

Calls a dialog to create a new inspection point.



Edit

Calls a dialog to edit an inspection point.



Complete Inspection Point

Calls a dialog to complete the inspection point. The suggested values for the quality decisions based on the inspection results are pre assigned in the corresponding configuration and can be altered.

Category Settings



Save

The user application settings, such as the columns, categories and their respective sizes as well as the size and display location of the application are saved after confirmation by the user. The user must confirm by entering "Yes" to the confirmation prompt.

Category Help



Help on Operation

This function key is used to call help on operating MOC. The base document is called "[moc_cc.pdf](#)". It contains the general instructions on using MOC and is valid across applications.



Help on Application

This function is used to call the application manual to access the help functions. The application manual contains the application functionality ordered along MES functionality and contains explanations on the expected information. The document also contains all detail applications of the application.



Help on Detail Application

This function calls the application manual at the bookmark where the detail application is described.

7.4 Selection Parameters

The application provides the following selection criteria.

Inspection Requirement Index Tab

- Area:
Selection menu containing all available areas, e.g. QM production
- Inspection requirement number:
Unique inspection requirement number

- Status:
The statuses to be used in the filter can be selected from a selection menu
- Plant:
SAP plant details
- Inspection batch
SAP inspection batch number
- Create PPS from:
Creation date from PPS, date can be selected
- Create PPS to:
Creation date from PPS, date can be selected

Order Index Tab

- Order number:
Associated SAP production order number
- Batch:
SAP batch number
- Origin:
from SAP
- Inspection type:
Selection menu containing inspection types, e.g. in-production inspection
- Purchase document number:
from SAP
- Material document number:
from SAP

Article Index Tab

- Article number
Material number
- Article designation
Material designation

- Customer material number
Customer's material number
- Customer material designation
Customer's material designation

Companies Index Tab

- Customer
Customer number
- Customer designation
- Supplier
Supplier number
- Supplier designation
- Manufacturer
Manufacturer number
- Manufacturer designation

User Fields Index Tab

- If user fields are defined during customization, this index tab allows filtering by the contents of the defined user fields.

Long Term Data Index Tab

- This filter allows the inclusion of long term data in the selection criteria.

All filter fields are ANDed together.

7.5 Master-Detail Grid (MD Grid)

The data is presented hierarchically in a Master-Detail Grid. To access the next lower level, click on the node marked by a "+" sign. As long as the data record contains multiple lower levels, these are each displayed in their own index tab. Depending on the detail application one of these index tabs is activated by default. The corresponding data from the MD Grid are displayed as a list in the active index tab. Clicking an index tab will change the list correspondingly. As long as the displayed list data can contain further lower levels, these contain a node marked with a "+" sign to access the next lower level.

Depending on the level and the active index tab in the MD Grid, the tool bar will activate the corresponding index tab containing the functions of the respective detail application.

A Master-Detail Grid displays the inspection requirements and all related data records. The first level of the MD Grid displays the inspection requirements (according to the inspection batch). The second level displays the inspection steps of the inspection requirement (e.g. the process steps, operations). The third level contains the inspection points for the respective inspection step. The next level contains the characteristics of the inspection points.

When no inspection points are used, the characteristics can be found directly below the inspection steps. In this case, the characteristics of the inspection steps are displayed.

MD Grid Structure

- Inspection Requirements
 - Inspection Steps
 - Inspection Points
 - Characteristics
 - Characteristics

7.6 Detail Application Inspection Requirement

This application displays the inspection requirements (inspection batches). These are uniquely identified by the inspection requirement number. Using the criteria from the selection panel will limit the amount of data displayed. Further filtering is possible through standard filters within the list.

It is not possible to edit or create inspection requirements since the data is transferred to the MOC through the interface only. Usage decisions are an exception. The tool bar contains a function button to start a new detail application for a usage decision for an inspection requirement that has not been completed yet.

The detail data of an inspection requirement are split up in the index tabs

- Inspection Requirement
- Companies
- Status - Details
- Documents
- Administration and
- User Fields (if defined).

Within these index tabs the data are grouped according to contents and functional aspects. Since no details can be modified (with the exception of usage decisions), a detailed description of the individual fields is not presented here. The groups in each index tab are described below.

Inspection Requirement Index Tab

This index tab is divided into the groups

- Identification
- Details and
- Article.

The group "Identification" contains all relevant order information. This includes the unique inspection requirement number, the inspection batch, order and purchase order numbers, the batch and the plant. The Details index tab contains the information on quantities and storage location. The Article index tab contains the component.

Companies Index Tab

This index tab is divided into the groups

- Customer
- Supplier and
- Manufacturer

and contains the supplier's batch and material number in addition to the relevant company number and designation.

Status - Details Index Tab

This index tab is divided into the groups

- Ratings
- Inspection Plan
- Confirmation
- Code
- Catalog Type
- Selected Set and
- Plant.

The important information in this index tab is the status (e.g. created, partial result, finished), inspection plan type and version and the usage decision information.

Documents Index Tab

This index tab contains information on purchase document number and position, material document number and position and document posting date along with other information.

Administration Index Tab

This index tab contains details on date, time and persons used for example to create, close and modify a data record.

User Fields Index Tab

This index tab is only visible when user fields have been defined for this application.

7.7 Detail Application Usage Decision

When an inspection requirement has not been completed yet, the usage decision (if configured accordingly) can be defined and the inspection requirement can be completed by clicking the button "Complete". The available usage decisions are stored in the PPS system and can be selected.

The usage decision cannot be modified for inspection requirements that are complete. After taking a usage decision of an inspection requirement, all associated data are uploaded to the PPS system (e.g. SAP).

7.8 Detail Application Inspection Step

This application displays the inspection steps of the selected inspection requirement (inspection batch). The inspection steps generally match the operations of the PPS system (e.g. SAP). Depending on the MOC configuration or in the case of linked operations in SAP-QM, an inspection step can also match a part of an operation. For example, the characteristics of an operation to be inspected can be split across multiple inspection stations where each inspection station is presented as a separate inspection step. In this case, the inspection step only contains those characteristics that can be inspected at the corresponding inspection station.

It is not possible to edit, create or delete inspection steps since the data is transferred to the MOC through the interface only.

Since in MOC every inspection step is associated with an operation, the completion of an inspection step is performed by logging off from the corresponding operation. Re-releasing an inspection step (e.g. for a subsequent inspection) can be accomplished by reactivating the finished operation. Such re-release is possible only when the corresponding inspection requirement has not been completed yet.

The detail data of an inspection step are split up in the index tabs

- Inspection Step
- General Inspection Point
- Inspection Point Evaluation
- Inspection Point Identification
- ADE Operation Order Data
- Inspection Step Order Data
- Confirmation
- Administration
- User Fields (if defined).

Within these index tabs the data are grouped according to contents and functional aspects. Since no details can be modified, a detailed description of the individual fields is not presented here. The groups in each index tab are described below.

Inspection Step Index Tab

This index tab is divided into the groups

- Identification
- Rating
- Workplace and
- Details.

The Identification group contains the unique inspection step number assigned by the MOC, the Rating group contains the status (e.g. created, partial result, finished) and the result (e.g. correct, fail). The Details group contains the plant, the cost center, the storage location and the subsystem ID.

General Inspection Point Index Tab

This index tab is divided into the groups

- Cycle and
- Properties.

The Cycle group includes whether the inspection interval is time or piece related. In this case, the unit and interval are also included.

Inspection Point Evaluation Index Tab

This index tab is divided into the groups

- Details
- Suggestion, Pass
- Suggestion, Fail and
- Property.

Properties are e.g. whether the evaluation of an inspection point is required. The suggestion groups display which details pass or fail the evaluation.

Inspection Point Identification Index Tab

This index tab is divided into the groups

- Details and
- User Fields.

The configuration of the PPS system determines which fields can later be used in the inspection points and what the user fields actually stand for. This index tab also specifies which fields are required.

ADE Operation Order Data and Inspection Step Order Data Index Tabs

These index tabs are not subdivided into groups and contain identical field designations.

Confirmation Index Tab

This index tab is organized into three confirmation categories and contains details on date and time of the respective confirmation.

Administration Index Tab

This index tab contains details on date, time and persons used for example to create, close and modify a data record.

User Fields Index Tab

This index tab is only visible when user fields have been defined for this application.

7.9 Detail Application Inspection Points

This application displays the inspection point details of the selected inspection point. If an inspection does not contain any inspection points, this detail application is absent.

When defining an inspection point, only some the fields of the index tabs Identification and Quantities can be entered. The leading system determines which fields can be entered. Changing inspection points functions similarly. Key fields used to identify the data record and specifications predefined by the leading system cannot be edited at all.

The tool bar of the maintenance dialog in the Evaluation index tab contains a button Complete to complete the inspection point. Default values are set according to the configuration and the inspection results.

The detail data of an inspection step are split up in the index tabs

- Inspection Point
- Details
- Identification
- Quantities
- Evaluation
- Confirmation
- Administration
- User Fields (if defined).

Within these index tabs the data are grouped according to contents and functional aspects. Since only few details can be modified, a detailed description of the individual fields is not presented here. The groups in each index tab are described below.

Inspection Point Index Tab

This index tab is divided into the groups

- Area
- Identification and
- Rating.

The Identification area contains the criteria that define the unique key. In addition to the inspection requirement and inspection step number and the inspection batch, these include the unique inspection point number. The Rating group contains the inspection point status (e.g. open, finished).

Details Index Tab

This index tab is not subdivided. It includes the indication whether the Inspection point is generated based on a time or piece interval or freely.

Identification Index Tab

This index tab is divided into the groups

- Details and
- User Fields.

The contents of these fields can be altered depending on the configuration and the defaults from the leading system. The field designations are also determined by the leading system and can vary.

Quantity Index Tab

The Quantity index tab is not subdivided and is used to enter the scrap and rework details.

Confirmation Index Tab

This index tab is organized into three confirmation categories and contains details on date and time of the respective confirmation.

Evaluation Index Tab

This index tab contains details defined and set during the completion of the Inspection point.

Administration Index Tab

This index tab contains details on date, time and persons used for example to create, close and modify a data record.

User Fields Index Tab

This index tab is only visible when user fields have been defined for this application.

7.10 Detail Application Characteristics

This detail application is located at the level immediately below the Inspection steps and Inspection points if Inspection points are used.

Details are shown about the Inspection characteristics highlighted in the list of characteristics; it is not possible to make modifications to the characteristics. Additionally, it is not possible to insert new characteristics or delete existing ones. Many of the data displayed here determine the inspection result and some can be viewed during the inspection.

The detail data of a characteristic are split up in the index tabs

- Characteristic

- Inspection Method
- Details
- Specifications
- Chart 1
- Chart 2
- Evaluation
- Confirmation
- Administration
- User Fields (if defined).

Within these index tabs the data are grouped according to contents and functional aspects. Since no details can be modified (with the exception of usage decisions), a detailed description of the individual fields is not given here. The groups in each index tab are described below.

Characteristic Index Tab

This index tab is divided into the groups

- Identification
- Properties and
- Weighting.

The identification area uses the number, the designation and the unique OP identification (work plan number) to describe the characteristic. The inspections are ordered by OP. Once a characteristic has been inspected, the next characteristic in sequence is shown and presented for inspection.

The properties contain the type of characteristic (e.g. measured sample value, sample evaluation), the type of evaluation (e.g. number of fields), the recording method of the result (e.g. must be tested) and a test schedule including frequency.

Testing Method Index Tab

This index tab is not subdivided and describes the inspection method proper, including version information.

Details Index Tab

This index tab is divided into the groups

- Gage and
- Additional Information.

The gage data displayed (number and designation) are not part of the QM-IDI because the QM-IDI does not transfer any gage data. This information is required to connect to a gage and to collect data on line and must be transferred by customer specific additional functions. The Additional information group contains the three QM-IDI category user fields plus an information field.

Specifications Index Tab

This index tab is divided into the groups

- Sampling Scheme
- Constructional Measures and
- Other Limit Values.

The Sampling Scheme group contains the inspection scope (e.g. may be smaller, must be followed exactly), the sampling unit and the rejection and acceptance quantity. The Constructional Measures group contains the tolerances and plausibility limits. Further limits displayed include the data "first / second upper / lower limit".

Chart 1 and Chart 2 Index Tabs

The action or warning limits defined for the respective control chart (e.g. Xq) are shown.

It is important to note that none of the fields in these two index tabs are part of the standard functionality of the QM-IDI interface. The corresponding data must be provided optionally through user specific additional functions.

Confirmation Index Tab

This index tab is organized into three confirmation categories and contains details on date and time of the respective confirmation.

Evaluation Index Tab

This index tab contains the details Site, Catalog Type and Selected Set for Evaluation 1 through 5.

Administration Index Tab

This index tab contains details on date, time and persons used for example to create and modify a data record.

User Fields Index Tab

This index tab is only visible when user fields have been defined for this application.

7.11 Detail Application Errors

In a separate detail application the defects documented in the inspection process (error type and location) are displayed both as a list and graphically. These data are displayed in an index tab integrated in the detail application Inspection Requirement. This ensures that the list contains all defects recorded in the inspection process of this order, the operation of the order, an associated inspection point, characteristic, sample or measured value.

The list of errors visualize the key fields

- Inspection Step Number
- Operation Sequence Number
- Sample and
- Value Number.

plus the details

- on defect type (defect type and location)
- on characteristic number and designation
- on operation number and designation of the inspection step or order
- on work plan number and designation

- on weighting
- a comment
- on error date and time and
- the administration duration.

The processing can be customized through the various possibilities for grouping and filtering in this list.

The data display on the top 10 defects can be switched from a list to a bar or pie chart. The list of defects is the basis for these data. If the list of defects is filtered when displayed as a list, this same filtering will be applied when displaying the data graphically. For the graphical display, the user can choose whether to show absolute numbers or percentages.

7.12 Detail Applications Control Chart 1 + 2, Histogram, Single Values 1 + 2, Statistics and Inspection Data

The detail applications

- Control Chart 1,
- Control Chart 2,
- Histogram and
- Statistics / Inspection Data
- Inspection Data

are integrated into the detail application Characteristics as additional index tabs. The Control chart index tabs display the characteristics data in the Control Chart 1 and 2 format by default. If the characteristic does not contain any control chart defaults, the data for a variable characteristic will be taken from Xq and s charts and those for an attributive characteristic will be taken from p and u charts.

Both index tabs for individual values display the contents of the respective control chart as a list. Sample control charts display all statistics values on the sample, including date, time and person responsible for the completion of the sample.

The statistics index tab includes details on number of samples and their measured values, maximum and minimum values and process parameters Cp and Cpk.

The Inspection data index tabs lists the inspection data in the finest granularity (e.g. the measured values level) independent of the selected control chart. This includes all associated key information (Inspection requirement, inspection step etc.). The contents of the list can be tailored through the sort, group and filter functions.

The customization settings determine the base data. The standard configuration displays all data that refer to the selected OP number and the associated Inspection requirement and Inspection step. If this configuration has been modified during customization, the base data can be manually and temporarily changed to refer to the data of the respective inspection step or of the selected characteristic (OP number).